

# Lab in the Cloud for IoN-DRI @ UCL

Tibor Auer



## Scientific Computing

### Challenge

- Research labs' needs for computing resources (hardware, software, users) change according to grants and recruitment
- Researchers require independent and decentralised solutions
- Research excellence requires reproducible solutions

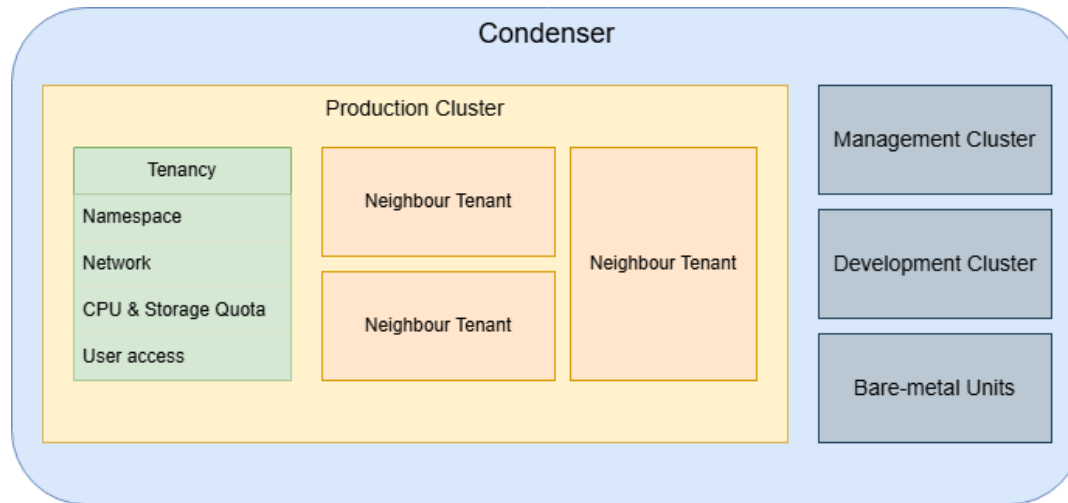
**How can we provide  
dynamic and heterogeneous services  
efficiently and sustainably?**



# PaaS for Scientific Computing

## Cloud-native approach

- Enables to dynamically provision computing resources → **scalability and sustainability**
- [Cloud @ UCL – Condenser](#)



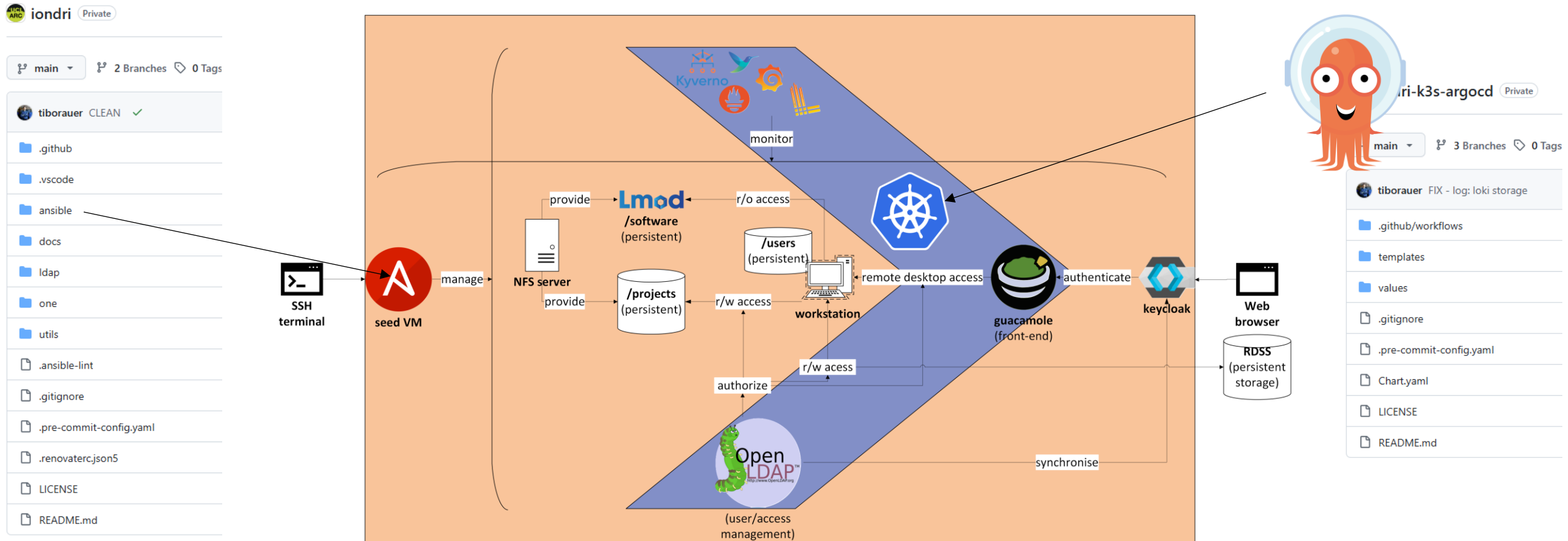
## GitOps

- Enable to use code as a single source of truth for infrastructure → **automatization and reproducibility**



# Lab in the Cloud

## Implementation for IoN-DRI @ UCL



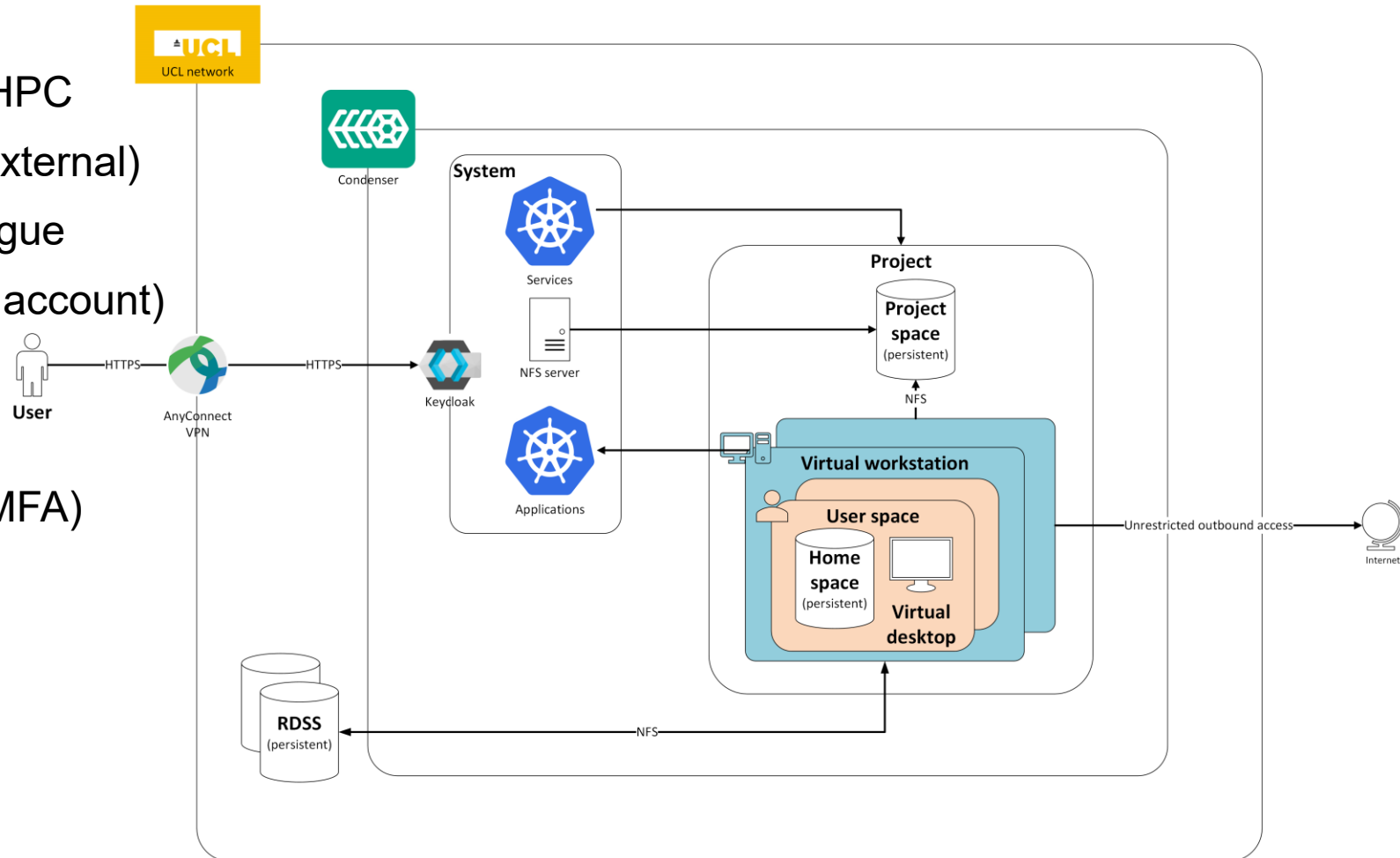
# Lab in the Cloud

## Current state

- Versatile resources: workstations, K8s, HPC
- Access to storage (within-platform and external)
- Pre-configured research software catalogue
- Accessible from web browser (with UCL account)

## Future

- Security hardening (e.g., krb5s, mTLS, MFA)
- ISMS coverage for sensitive data (CRE)
- Automated and scalable data pipelines



# Lab in the Cloud

## Front-ends

- Documentation on SharePoint:  
[IoN-DRI Computing Platform](#)
- User access via web browser:  
[Workstations portal](#)  
[Self-service password portal](#)
- Maintenance  
[LDAP Admin portal](#)  
[Monitoring portal](#)  
[ArgoCD](#)  
[K8s dashboard \(Headlamp\)](#)  
[Alerts on Slack](#)



## IoN-DRI Computing Platform

 **Tibor Auer**  
Senior Research Infrastructure Developer (Faculty of Brain Sciences)

### Overview

We are developing the IoN-DRI Computing Platform, which will be available for IoN-DRI researchers, as part of the Scientific Platform which is being developed for the IoN-DRI programme.  
The IoN-DRI Computing Platform provides computing resources tailored to various specific research needs and ensures the robustness and scalability of these computing resources. These resources can include workstations with desktop, application servers, as well as, clusters for high-performance computing, Kubernetes, and more.

### Maintainers

 **Tibor Auer**  
Senior Research Infrastructure Developer (Faculty of Brain...

### Resources

#### Workstations

- Independent machines with dedicated resources (CPU, RAM, disk)
- Desktop for interactive work

#### Software

- Docker/Apptainer
- Python (miniforge)
- VS Code
- MATLAB
- OCTAVE
- FreeSurfer
- FSL
- dcm2niix
- DCMTK

#### Storage

- User space (machine-specific)
- Collaborative project space shared across the machines
- Mounted RDSS

### Access

Please note, that access is only available via the UCL campus network or through the UCL VPN.

Please, contact one of the [maintainers](#) to register your UCL username and provide with an (initial) password for the platform. Then, you can use these credentials or **your UCL account** to access the links below.

### Platform

 **Self-service Password Portal**  
(only with the platform password)

 **Workstations**  
(also with UCL account)